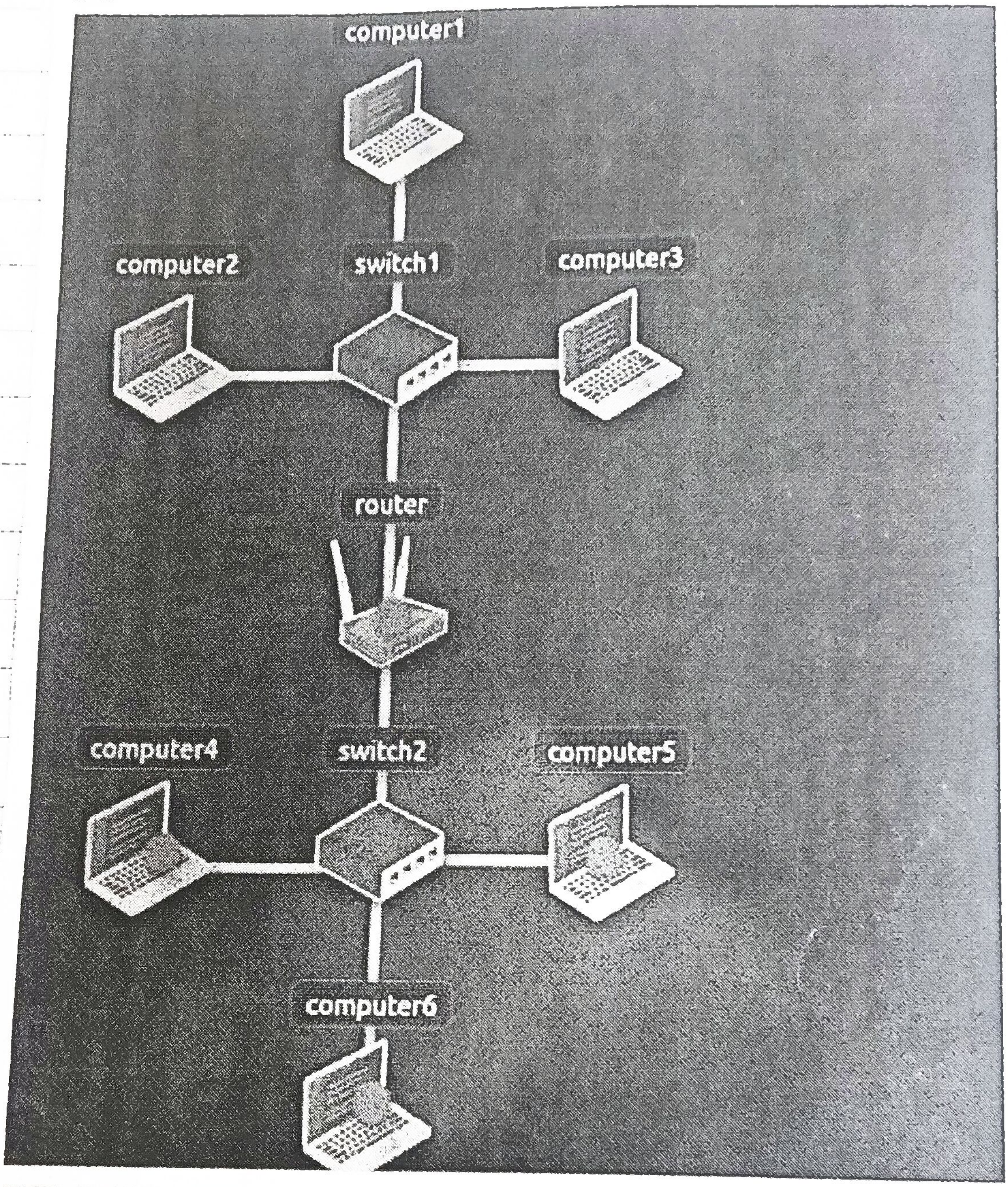
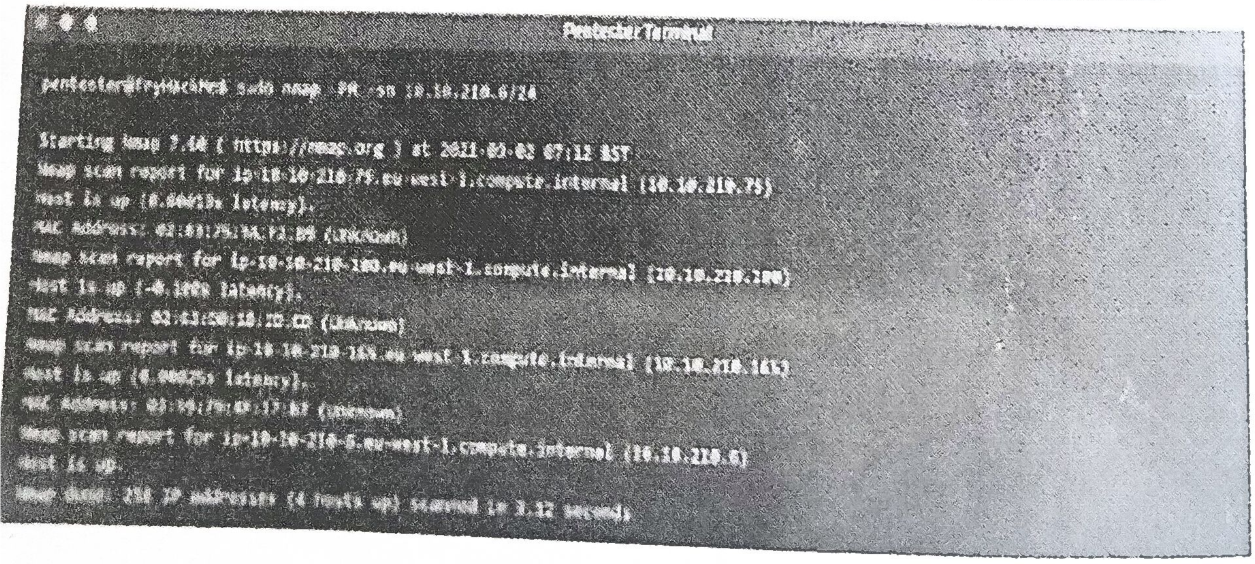
8-a





```
pentester@TryHackMe$ sudo nmap -PR -sn 10.10.210.6/24

Starting Nmap 7.60 ( https://nmap.org ) at 2021-02-02 07:12 BST
Nmap scan report for ip-10-10-210-75.eu-west-1.compute.internal (10.10.210.75)
Host is up (0.00013s latency).
MAC Address: 02:83:75:3A:F2:89 (Unknown)
Nmap scan report for ip-10-10-210-100.eu-west-1.compute.internal (10.10.210.100)
Host is up (-0.100s latency).
MAC Address: 02:63:D0:1B:2D:CD (Unknown)
Nmap scan report for ip-10-10-210-165.eu-west-1.compute.internal (10.10.210.165)
Host is up (0.00025s latency).
MAC Address: 02:59:79:4F:17:B7 (Unknown)
Nmap scan report for ip-10-10-210-6.eu-west-1.compute.internal (10.10.210.6)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.12 seconds
```

0/9/26

Ex 08. NMAP(8) Discover Live Hosts using NMap Scans ( ARP, ICMP, TCP/UCP) on TryHackMe Platform.

(a)

Aim:

To discover Live Hosts using Nmap Scans ( ARP, ICMP, TCP/UDP) on the TryHackMe Platform Room link: https://tryhackme.com/Nmap01

① Introduction:

NMAP is free and open source network scanning tool used to discover action system and Identify running services on Network. Scanning offline System wastes time and can create unnecessary network traffic

② Subnetworks:

A network segment is a group of Computers connected through a common Medium such as Ethernet switch/WI-FI access Point

eg

| Subnet | Subnet Mask | Approx. Hosts |
|---|---|---|
| /16 | 255.255.0.0 | ~65,000 |
| /24 | 255.255.255.0 | ~250 . |

3) Enumerating target.

* Before Scanning, NMAP needs to know which system / IP addresses should be checked.

targets

→ Single IP address : 10.10.10.5
→ Range : 10.10.10.5-10
→ Subnet : 10.10.10.0/24
→ Display targets without Scanning : NMAP -SL targets

4) Live Host Discovery:

The Experiment covers 4 Protocols:

* ARP - Link layer
* ICMP - Network layer
* TCP - Trogsort layer.
* UDP - Transport layer.

5) ARP Host Discovery:

* ARP requests are used to find Live Host on some subnet.
* Alive host responds with an ARP reply
* ARP cannot Cross routers
* In the tryHackMe activity, 3 devices were discovered.

6) ICMP Host Discovery:

* It uses 8 types (Echo Request) and type 0
* NMAP option : -PE.
* ICMP timestamp scan : -PP.
* ICMP Address Mask scan : -PM.

7) TCP SYN Ping:

- Sends a TCP Packet with SYN flag set.
- An open Port May respond with SYN/ACK.
- NMap OPtion: -PS.

Eg- Telnet Port 23 - RI23

⑧ TCP Ack Ping:

- Sends a TCP Packet with the Ack flag set
- Normally uses Port 80 by default.
- NMap option: -PA.
- A target may respond with RST, Indicating that host is active.

⑨ UDP Ping:

- UDP Packets can be used to determine whether a host in active.
- A Packet sent to a closed UDP Port May Procedure an ICMP "Port unreachable"

⑩ Reverse DNS Lookup:

- NMap Manually Performs reverse DNS lookups for online hosts.
- -n → disables DNS resolution
- -R → performs reverse DNS.

Result: The NMap live Host Discovery experiment was successfully studied on TryHack platform. Live host were Identified using ARP, ICMP, TCP, UDP techniques and corresponding NMap were understood.